

WEEK 15

Week-15-Pointers: Attempt review | REC-CIS - Personal - Microsoft Edge

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REC-CIS

Correct
Marked out of 1.00
Flag question

Example
 $arr = [1, 3, 2, 4, 5]$
Return the array $[5, 4, 2, 3, 1]$ which is the reverse of the input array.

Function Description
Complete the function `reverseArray` in the editor below.
`reverseArray` has the following parameter(s):
`int arr[n]`: an array of integers
Return
`int[n]`: the array in reverse order

Constraints
 $1 \leq n \leq 100$
 $0 < arr[i] \leq 100$

Input Format For Custom Testing
The first line contains an integer, n , the number of elements in arr .
Each line i of the n subsequent lines (where $0 \leq i < n$) contains an integer, $arr[i]$.

Sample Case 0
Sample Input For Custom Testing
5
1
3
2
4
5

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REC-CIS

```
28 *      *(a + 1) = i + 1;
29 *    }
30 *
31 *    return a;
32 *
33 *
34 */
35 int* reverseArray(int arr_count, int *arr, int *result_count) {
36     *result_count = arr_count;
37     int *reversedarr = (int*) malloc(arr_count * sizeof(int));
38     int *start = arr;
39     int *end = arr + arr_count - 1;
40     while(start <= end){
41         *reversedarr++ = *end--;
42     }
43     return reversedarr - arr_count;
44 }
45
46
```

Test	Expected	Got	
✓ int arr[] = {1, 3, 2, 4, 5};	5	5	✓
int result_count;	4	4	
int* result = reverseArray(5, arr, &result_count);	2	2	
for (int i = 0; i < result_count; i++)	3	3	
printf("%d\n", *(result + i));	1	1	

Passed all tests! ✓

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Question 2
Correct
Marked out of 1.00
Flag question

An automated cutting machine is used to cut rods into segments. The cutting machine can only hold a rod of *minLength* or more, and it can only make one cut at a time. Given the array *lengths[]* representing the desired lengths of each segment, determine if it is possible to make the necessary cuts using this machine. The rod is marked into lengths already, in the order given.

Example

$n = 3$
 $lengths = [4, 3, 2]$
 $minLength = 7$

The rod is initially $sum(lengths) = 4 + 3 + 2 = 9$ units long. First cut off the segment of length $4 + 3 = 7$ leaving a rod $9 - 7 = 2$. Then check that the length 7 rod can be cut into segments of lengths 4 and 3. Since 7 is greater than or equal to $minLength = 7$, the final cut can be made. Return "Possible".

Example

$n = 3$
 $lengths = [4, 2, 3]$
 $minLength = 7$

The rod is initially $sum(lengths) = 4 + 2 + 3 = 9$ units long. In this case, the initial cut can be of length 4 or $4 + 2 = 6$. Regardless of the length of the first cut, the remaining piece will be shorter than $minLength$. Because $n - 1 = 2$ cuts cannot be made, the answer is "Impossible".

Function Description

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```
26 * }
27 *
28 */
29 char* cutThemAll(int lengths_count, long *lengths, long minLength) {
30     long t=0,i=1;
31     for(int i=0;i<lengths_count-1;i++){
32         t+=lengths[i];
33     }
34     if(t-lengths[lengths_count-1]<minLength){
35         return "Impossible";
36     }
37     while(i<lengths_count-1);
38     return "Possible";
39 }
40 }
41 }
```

	Test	Expected	Got	
✓	long lengths[] = {3, 5, 4, 3}; printf("%s", cutThemAll(4, lengths, 9))	Possible	Possible	✓
✓	long lengths[] = {5, 6, 2}; printf("%s", cutThemAll(3, lengths, 12))	Impossible	Impossible	✓

Passed all tests! ✓

Finish review